FIGNN: Fuzzy Inference-Guided Graph Neural Network for Fault Diagnosis in Industrial Processes

Youqing Wang, *Senior Member, IEEE*, Tengxiao Yin, Shenggang Zhu, Xin Ma, Li Liang, and Jipeng Guo, *Member, IEEE*

***Abstract*—Intelligent fault diagnosis in industrial systems is important for improving production efficiency. The complex interactions between sensor units could be represented as graph and are beneficial for identifying the operational status of industrial systems. Recently, graph neural networks (GNNs) have attracted widespread attention and achieved satisfactory performance in fault diagnosis. The existing GNN-based fault diagnosis methods with neighbor information aggregation and MLP classifier mainly extract features strongly correlated with labels, rather than causal features, limiting the reliability and interpretability of the diagnostic results. To address these challenges, this article proposes a fuzzy inference-guided GNN (FIGNN) for fault diagnosis. First, a fuzzy inference-guided information aggregation is skillfully designed to extract causal features by fuzzy rules, which is capable of distinguishing normal and fault characteristics. Then, the FIGNN introduces additional conclusion nodes as classifiers, achieving the explicit inference interactions between the final fault and sensor signals. Finally, to enhance the reliability and flexibility of FIGNN, a network pruning-based graph structure construction approach is developed, which effectively utilizes inference relationships to adaptively construct a more accurate graph. Experimental results on the Tennessee Eastman (TE) process and the three-phase flow process show the significant performance improvements of 4.18% and 0.81% and strong reliability of the proposed FIGNN in complex industrial fault diagnosis. The code could be available at https://github.com/SoarYin/FIGNN.git**

***Index Terms*—Fault diagnosis, fuzzy inference, graph neural network (GNN), high-dimensional industrial data, representation learning.**

I. Introduction

ANALYZING and representing complex sensor information is the basis for fault diagnosis. With the continuous improvement of the integration and integrity of industrial equipment and sensor networks, industrial process data exhibit increasingly complex and high-dimensional characteristics. Traditional manual fault diagnosis methods are difficult to process complex industrial data, which contains redundant features. Therefore, more advanced and comprehensive approaches are applied to fault diagnosis. These fault diagnosis approaches include three main categories: mechanism model-driven [1], knowledge-driven [2], and data-driven [3] approaches. Mechanism model-driven fault diagnosis methods are limited by the uncertainty of dynamic mechanism in the industrial production process, which limits its generality in real industrial applications [4].

Knowledge-driven fault diagnosis approaches provide an inherently interpretable way to analyze operational conditions in complex industrial processes. The combination of knowledge base (KB) and inference engine was uniformly represented by ontologies and analyzed by rules [5]. The data was linked with their domain metadata, and the diagnostic conclusion can be obtained from the constructed knowledge graph (KG) [6]. However, the representation ability of knowledge-driven methods heavily relies on large amounts of manual parameter tuning. To overcome this shortcoming, some machine learning optimization methods are embraced by knowledge-driven approaches. For instance, Straub [7] applied gradient descent and pruning methods as network parameter tuning and rule construction methods in a rule-fact expert system. Meanwhile, Meilicke et al. [8] proposed a bottom-up technique for effectively learning logical rules and uncertain rules from large KGs. Dettmers et al. [9] proposed a convolution over KG embedding representations to predict links in KG. These methods were proposed to compensate for the shortcomings of the knowledge-driven approach in terms of representation abilities, but an optimal KG construction is still a difficult task.

The advantage of data-driven approaches is their strong ability to feature extraction, which can effectively represent high-dimensional and complex industrial process data [10]. Sensor network data could be organized in a matrix form and then analyzed by a convolutional neural network [11]. Data-driven methods, especially deep convolutional neural networks (DCNNs), have shown better performance than traditional methods in various fields [12]. However, the translation invariance is a fundamental and necessary property of DCNNs [13], but sensor network data fundamentally does not satisfy it [14]. As a result of this mismatch, these methods cannot

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Youqing Wang, Tengxiao Yin, Xin Ma, Li Liang, and Jipeng Guo are with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: wang_youqing@mail.buct.edu.cn; tengxiaoyin@buct.edu.cn; 2023700002@buct.edu.cn; lianglibuct@buct.edu.cn; guojipeng@buct.edu.cn).

Shenggang Zhu is with the State Research and Development Center of Control System and Information Security Technologies for Energy Industry, CHN Energy Technology and Environment Limited, Beijing 100039, China (e-mail: zhshgh@aliyun.com).

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make use of intrinsic sample characteristics to achieve the discriminative data representation and forcibly rely on the nonlinear fitting ability of DCNNs to extract useful features. Some specialized convolutional network approaches have been used to achieve more effective representation learning. For example, applying 1D-CNN model to force CNN only extract temporal information [15]. Meanwhile, recurrent neural networks (RNNs) were specifically designed for time-series data fault diagnosis [16]. The most widely used RNN approaches are long short-term memory (LSTM) and gated recurrent unit (GRU) neural networks [17]. Only considering temporal relationship is imperfect, topological structure among sensors is also important, so some comprehensive methods combining CNN and LSTM layers could improve the performance [18].

In fact, temporal methods ignore the spatial structure information, which is important to sensor network analysis. Thus, accurately modeling the non-Euclidean relationships in spatial dimension is an important problem. In this regard, graph neural networks (GNNs) are highly suitable for non-Euclidean data [14], and they have attracted significant attention in fault diagnosis. There are two main technology routes for GNN approaches in fault diagnosis. First, the samples of sensor data matrix were treated as graph nodes v with signal x, and then information of similar samples was aggregated via graph Fourier transformation filter g [19]. The other way was to set some hidden states as node v [20], and the nodes are not homogeneous, which requires a different information aggregation way defined by a message passing function [21]. Specified information aggregation ways satisfied the requirement of representing and analyzing a complex network. Wu and Zhao [22] manually analyzed topology connections of process components and then constructed a graph for GNNs. Sun et al. [23] leveraged a multichannel residual network to better extract the comprehensive feature for GNNs. The method in [24] utilized trainable vectors to measure interactions between components and learned data embedding via interaction graph, and then attributed fault diagnosis to a graph classification problem. Jia et al. [25] initialized a topological graph using a flowchart and adaptively updated the graph via a self-attention mechanism. Wu et al. [26] proposed an iterative deep graph learning and embedding framework for semi-supervised fault diagnosis with limited labeled data.

Although the above GNN-based fault diagnosis methods have achieved certain success, some prior knowledge has been applied to graph construction and design of GNNs for fault diagnosis [27], [28], [29]. Li et al. [30] utilized Granger causality to construct a reliable graph for graph attention network (GAT) in time series forecasting. Causality relationships indicate the direction of feature aggregation. On this basis, Wang et al. [31] designed a double-head GAT that could avoid non-causality connections and provide reliable prediction results. Wang et al. [32] applied an optimized principal neighborhood aggregation to collect more features to diagnose weak faults. Different from implicit knowledge, Guo et al. [33] injected process knowledge into GNNs and used attention mechanisms to model the dependencies between sensors. For these graph network methods mentioned above, they try to build a better graph structure by analyzing the intrinsic mechanisms.

The GNNs in fault diagnosis have shown superior performance, which effectively fuses information from multiple industrial components by a neighboring message-passing strategy. However, due to complex interactions between sensor components and signals, it is enormously important to filter out the irrelevant features and capture crucial connections of the input graph. Most existing GNN-based fault diagnosis approaches utilize traditional information aggregation to extract features and then simply utilize MLPs as a final classifier to achieve fault diagnosis, so that they mainly focus on the "surface" information only strongly correlated with objective labels instead of extracting causal features truly suitable for decision-making of fault diagnosis. This inevitably weakens the reliability and interpretability of diagnostic results.

To understand the essence of decision-making in GNNs, several post hoc explanation methods have been proposed. Explanation by simplification is the most widely used technique under the category of independent post-modeling explanation techniques. Among them, the local interpretable model-agnostic explanation (LIME) [34] was the most representative method and also available on GNNs [35]. Another approach to explanation is feature relevance explanation, and Shapley additive explanation (SHAP) [36] is one of the most representative methods, which assigns each feature an importance value for a particular prediction. These approaches fall short in integrating relational information, which is the essence of the graph. Therefore, GnnExplianer [37], which takes advantage of the rich relational information provided by the graph and the node characteristics, captures the important graph structure and features for decision-making. Because they are post hoc explanatory mechanisms, they cannot improve the reliability and interpretability of diagnostic results through discriminative representation learning and direct diagnostic reasoning.

To address the above issues, this article develops a fuzzy inference-guided GNN (FIGNN) for industrial fault diagnosis. This method promotes the learning of causal features that are important for diagnosis by fuzzy inference-guided information aggregation. Then, the conclusion nodes are introduced into the graph structure as a fault diagnosis classifier, and thus, FIGNN could establish direct correlation interaction and achieve stable inference between diagnostic results and sensor signals. The main contributions of this study are as follows.

1) To enhance the reliability of diagnostic results, a novel information aggregation mechanism with Takagi–Sugeno fuzzy inference guided is proposed for FIGNN, which could find the causal features that significantly affect the diagnostic results by fuzzy rules.
2) To enhance the interpretability and traceability of diagnostic results, FIGNN introduces the conclusion nodes in the graph as diagnostic classifiers and achieves direct inference interaction between sensor components in industrial systems and diagnostic results.

3) A novel and flexible graph structure construction approach with pruning and inference functions is developed, which adaptively captures more accurate and essential interactions not only between sensors but also between sensors and conclusion nodes.

The remainder of this article is organized as follows. Section II reviews the fault diagnosis problem and some relevant preliminaries (including graph convolutional network (GCN) and the adaptive network-based fuzzy inference system). Section III provides detailed descriptions of the proposed FIGNN method. The experimental performance of FIGNN is evaluated and analyzed in Section IV. Finally, Section V concludes the entire paper.

II. Preliminaries

A. Fault Diagnosis Using Deep Learning Approaches

The working condition of an industrial manufacturing system is controlled by different controllers, with information collected by different sensors. The behavior of each component is recorded as a time series $\mathbf{S}_i = [S_{i1}, S_{i2}, \ldots, S_{it}]$. These data are collected by a control system with n components and form a history data matrix $\mathbf{S} = [\mathbf{S}_1^T, \mathbf{S}_2^T, \ldots, \mathbf{S}_n^T]$, which is used for fault diagnosis problems. With a long-time span, the history data matrix $\mathbf{S}$ usually needs to be divided into many segments by sliding window methods. The final sample is formed as $\mathbf{S}_{\text{samp}} = [\mathbf{S}_1^T, \ldots, \mathbf{S}_i^T, \ldots, \mathbf{S}_n^T]$, where $\mathbf{S}_i = [S_{it}, S_{(it+1)}, \ldots, S_{(it+w)}]$. Due to the high-dimensional and complex characteristics of industrial data, deep learning-based fault diagnosis methods utilize multilayer neural networks to extract discriminative features for fault diagnosis. Existing methods usually formulate the fault diagnosis problem as a classification task and utilize the MLPs as a classifier to predict labels. This leads to the feature extractor overly focusing on "surface" features that are only correlated with labels, rather than the essential features that are truly useful for fault diagnosis. This can easily limit the reliability and traceability of diagnostic results.

B. Graph Convolutional Network

Here, we provide some preliminaries, including the definition of graph, GCN, and neural network pruning for GCN.

1) Definition of Graph: A graph is the dominant way to model non-Euclidean relationships in sensor networks. A graph is denoted by $\mathcal{G} = \{X, \mathcal{V}, \mathcal{E}\}$, where $\mathcal{V} = \{v_1, v_2, \ldots, v_n\}$ represents nodes set with n nodes, $\mathcal{E}$ denotes edges set of the graph $\mathcal{G}$, and $X \in \mathbb{R}^{n\times d}$ is the attribute feature of graph signal nodes. The adjacency matrix $A \in \mathbb{R}^{n\times n}$ is used to describe the topological structure of the graph $\mathcal{G}$, where $A_{ij} = 1$ if nodes v_i and v_j are connected (i.e., $(v_i, v_j) \in \mathcal{E}$), otherwise, $A_{ij} = 0$. Further, its normalized Laplace matrix is calculated using $L = I_n - D^{-1/2}AD^{-1/2}$, where D is the diagonal degree matrix with $D_{ii} = \sum_{j=1}^{n} A_{ij}$ and I_n is an n identity matrix. The graph $\mathcal{G}$ is an undirected graph if and only if A is a symmetric matrix, i.e., $A_{ij} = A_{ji}$. A graph is directed if its adjacency matrix A exists, where $A_{ij} \neq A_{ji}$. A graph where the adjacency matrix takes real numbers $A_{ij} \in \mathbb{R}$ is called a weighted graph. To model topological relationships between sensors more accurately, a directed graph is utilized to describe the topological structure in the proposed FIGNN method.

2) Revisiting GCN: To obtain the discriminative representation of graph data, GCN generalizes the convolution operation to the non-Euclidean space and effectively aggregates neighborhood information. There are two classic versions of GCN: spectral-based and spatial-based GCN. Since most GNN-based industrial fault diagnosis methods usually use sensors as nodes of the graph, the spatial graph networks are suitable and widely utilized. Spatial-based GCN performs information aggregation directly based on the topological structure of the graph. The graph convolutional operation of spatial-based GCN can be formulated as

$$m_i^{t+1} = \sum_{j\in\mathcal{N}(i)} I_{\text{ag}}\left(V_i^t, V_j^t\right) \tag{1}$$

$$V_i^{t+1} = U_{\text{tr}}\left(V_i^t, m_i^{t+1}\right) \tag{2}$$

where I_{ag} is the aggregation function used to collect information from neighboring nodes $\mathcal{N}(i)$, and U_{tr} is the transformation function to update the representation of node v_i in the tth iteration. In most spatial-based approaches, the aggregation and transformation functions are the same as those of spectral-based GCN by specifying I_{ag} and U_{tr}

$$m_i^{t+1} = \sum_{j\in\mathcal{N}(i)} I_{\text{ag}}\left(V_i^t, V_j^t\right) = \sum_{j\in\mathcal{N}(i)} D_{ii}^{-\frac{1}{2}} A_{ij} D_{ii}^{-\frac{1}{2}} V_j^t \Theta \tag{3}$$

$$V_i^{t+1} = \sigma\left(m_i^{t+1}\right). \tag{4}$$

The essence of these two approaches is different. Spectral-based GCN is suitable for neighbors without well-defined relationships. However, spatial-based GCN can describe some specified relationships by formula (1), which is suitable for nodes with obvious topological relationships like sensor networks. The features transformation by simple matrix multiplication $(D_{ii}^{-(1/2)} A_{ij} D_{ii}^{-(1/2)} V_j^t)\cdot\Theta$ focuses too much on features that are beneficial for correct label, which cannot effectively distinguish normal signals and abnormal signals and activate signals belong to the key causal parts for fault diagnosis. In this study, a novel information aggregation, which is different from (3), is defined according to an adaptive network-based fuzzy inference system (ANFIS) within the framework of spatial-based GCN. The fuzzy inference with fuzzy rules is capable of capturing the essential features that affect the diagnostic results.

3) GNN Pruning: The purpose of network pruning is to remove redundant connections, which reduces the complexity while maintaining the performance of the model. Judging the importance of connections is the same as the essence of network pruning and graph construction; thus, pruning methods can be applied in a wider range of applications. Given a dataset $\{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$ with a sample feature $\mathbf{x}_i$ and a truth label $\mathbf{y}_i$, network pruning can be formulated as an optimization problem with a given sparsity level κ, i.e.,

$$\min_{\mathbf{w}} \frac{1}{n} \sum_{i=1}^{n} \ell\left((\mathbf{x}_i, \mathbf{y}_i); \mathbf{w}\right)$$
$$\text{s.t. } \mathbf{w} \in \mathbb{R}^m, \|\mathbf{w}\|_0 \leq \kappa \tag{5}$$

where ℓ is the loss function, $\mathbf{w}$ represents network parameters, and $\|\mathbf{w}\|_0$ is the l_0 norm of $\mathbf{w}$, indicating the number of non-zero elements in the vector. Chen et al. [38] presented a unified framework that simultaneously prunes the graph adjacency matrix and the model, degrading the optimization of graph structure into a pruning problem. In their study [38], a loss-based iterative method is utilized to optimize parameters. However, this optimization problem is not efficiently solved by the iterative method because of its high complexity and the requirement of heavy hyperparameter tuning.

Metrics-based methods prune networks by measuring the contribution of parameters $\mathbf{w}$ to the training objectives. Both theoretical analysis and trial-and-error approaches can be applied. Meanwhile, magnitude-based methods are the most intuitive approaches because they determine the importance of connections using the magnitude of the parameter weights (i.e., $|\mathbf{w}|$). Nevertheless, weights are influenced by various scales of input data, which means that the magnitude of weights cannot accurately reflects their contributions. In comparison, saliency-based methods mainly use gradient-related information to form a scoring matrix $s(\mathbf{w}) = |(\partial\ell/\partial\mathbf{w}) \odot \mathbf{w}|$ [39]. Saliency-based methods, which are the dominant approaches, distinguish the importance of parameters and remove irrelevant parameters according to the saliency score until the target sparsity is reached.

C. Adaptive Network-Based Fuzzy Inference System

ANFIS is a combination of artificial neural network (ANN) and fuzzy inference theory, which could extract more fundamental features by fuzzy rules. Given the raw input data x and y, ANFIS learns the interpretable representation using a five-layer network. Layer 1 with premise membership function parameters achieves the fuzzification of input data, which fuzzifies x and y as membership degrees $[x_1, x_2]$ and $[y_1, y_2]$. Layer 2 determines the activation weight $[w_1, w_2]$ of the corresponding if-then rules according to the satisfaction of the rule premise. Layer 3 normalizes the weights. Layer 4 executes the inference function by utilizing membership degrees on the corresponding rule. The Takagi–Sugeno model is used to represent if-then rules, with an example shown below. Given two if-then rules, if x is x_1 and y is y_1, then z is z_1, and if x is x_2 and y is y_2, then z is z_2, as formulated in the following:

$$\begin{aligned} z_1 &= P_1(x) + Q_1(y) + c_1 \\ z_2 &= P_2(x) + Q_2(y) + c_2 \end{aligned} \tag{6}$$

where $P_i(x)$ and $Q_i(y)$ are polynomials in x and y, respectively. If $P_i(\cdot)$ and $Q_i(\cdot)$ are first-order polynomials, the inference result is a linear combination of membership degrees, which is similar to the T-norm operation "AND." The parameters in $P_i(\cdot)$, $Q_i(\cdot)$, and bias c_i in ANFIS are trainable; thus, they do not rely solely on expert knowledge. Finally, Layer 5 computes the overall output by adding together the outputs of Layer 4 with the normalized activation weight w_i, that is,

$$z = \frac{w_1}{w_1 + w_2} * z_1 + \frac{w_2}{w_1 + w_2} * z_2. \tag{7}$$

This article would analyze and show how this form of inference can be expanded to update the parameters of spatial-based GCN in Section III.

III. Proposed Method

In this section, to improve the reliability and interpretability of fault diagnosis results, an FIGNN is proposed, extending the fuzzy inference theorem to GNN. Here, the proposed FIGNN is introduced in detail from the following three aspects: 1) the main framework of FIGNN; 2) reliable graph structure construction for FIGNN; and 3) the traceability mechanism design of diagnostic results for FIGNN.

A. Fuzzy Inference-Guided GNN

Here, the details of FIGNN are described, including the modeling of the industrial process with a graph, fuzzy inference with information aggregation on the graph, and the training strategy of FIGNN.

1) Modeling the Industrial Process With a Graph: When GNNs are used for fault diagnosis in industrial processes, a crucial challenge is the interpretability of the readout and the corresponding decision-making functions, such as the MLP as a classifier. To improve reliability and traceability of diagnostic results, the final diagnostic results should also be obtained through inference, rather than simple MLP classifier. Predicates are used to represent both factual knowledge (e.g., states, properties, and concepts of things) and rule-based knowledge (e.g., causal relationships between things). Based on the theory of predicates, two kinds of nodes can be defined as follows.

a) Ground nodes: Ground nodes correspond to predicates that indicate factual knowledge. Here, we define ground nodes as predicates that can be directly inferred from a data sequence obtained by a single component in the industrial process. Specifically, the ground nodes can be expressed as

$$V_g = \mathrm{Concat}_{F_i \in F} M_i \in \mathbb{R}^{f*d} \tag{8}$$

where F is the fuzzy set determined by specific issues, and $f = |F|$ is the number of fuzzy sets. $M_i \in \mathbb{R}^d$ is the membership degree of the d-dimensional feature on the fuzzy rule $F_i \in F$. $\mathrm{Concat}(\cdot)$ indicates a concatenation operator, connecting different fuzzy sets to form a greater whole. The reason for using the concatenation operator is that parameters in the membership function are not shared between different features of a node. For ground nodes, features are provided by pre-processing and fuzzification operators.

b) Conclusion nodes: Conclusion nodes correspond to predicates that indicate rule-based knowledge. Here, we define conclusion nodes as predicates that require multiple preconditions from ground nodes to infer. A conclusion node can be determined by

$$V_c = P_c\left(\mathrm{Aggr}_{g \in \mathcal{N}(c)}\left(V_g\right)\right) \in \mathbb{R}^{n*1} \tag{9}$$

where $\mathrm{Aggr}(\cdot)$ is the information aggregation function, and $\mathcal{N}(c)$ is the neighboring nodes set of the conclusion node c. Both ground nodes and conclusion nodes can be included in a prediction function P_c that transforms aggregated features into prediction information. Different from ground nodes, conclusion nodes do not have the attribute of fuzzy sets, as each node represents the membership degree of only one fuzzy set, whereas defuzzification transforms the results into clear conclusions. Since the final fault class can be set to the conclusion node, there is no need for an MLP classifier to predict the diagnostic results from the fault features.

2) Information Aggregation With Fuzzy Inference on Graph:

a) Fuzzify layer: The first step of FIGNN is fuzzifying the input data to describe the fuzzy feature for ground nodes. The triangular membership function is widely used to describe the degree of membership of a feature to a fuzzy set. For any feature x, the triangular membership function of a fuzzy set is formulated as follows:

$$\mu(x) = \begin{cases} 0 & x < a \text{ or } x > c \\ \dfrac{x-a}{b-a} & a < x < b \\ \dfrac{c-x}{c-b} & b < x < c \end{cases} \tag{10}$$

where a, b, and c are trainable parameters correspond to the fuzzy set and feature, which are updated by the back–forward optimization of Layer 1 in FIGNN.

b) Graph fuzzy inference: In the classic ANFIS, an if-then rule is represented as a continuous formulation using first-order polynomials. Thus, we also simplify the inference process on a graph as follows:

$$y_z = \sum_{i=1}^{f*d} w_i V_g^i + c. \tag{11}$$

However, the above inference process (11) ignores the topological structure between nodes, inevitably reducing the reliability of the inference process. In fact, the adjacency matrix $A \in \mathbb{R}^{k\times k}$ describes important structural relationships between ground and conclusion nodes, where $k = f * d + n$. There are only two predicates, greater-than-normal and less-than-normal, in fault diagnosis, thus, the bias c is removed to simplify the model and reduce the training cost. The inference process can be written as a simple yet physical matrix multiplication form

$$\begin{bmatrix} V_g(t) \\ V_c(t) \end{bmatrix} = (\mathbf{w} \odot A) \begin{bmatrix} V_g(t-1) \\ V_c(t-1) \end{bmatrix} = \widehat{A} \begin{bmatrix} V_g(t-1) \\ V_c(t-1) \end{bmatrix} \tag{12}$$

where $V_g(t)$ and $V_c(t)$ are attribute features of the ground and conclusion nodes in the tth iteration, $V_g(0)$ is initialized by the triangular membership function $\mu(x)$, and $V_c(0) = \mathbf{0}$. Moreover, $\mathbf{w} \in \mathbb{R}^{k*k}$ is a learnable weight matrix in the inference process. In (12), the adjacency matrix A is a mask of the weight matrix $\mathbf{w}$, improving the discriminant of the inference process. Specifically, if $A_{ij} \neq 0$, there is a possibility of information aggregation and inference relationship among corresponding nodes. On the contrary, if $A_{ij} = 0$, FIGNN does not perform the inference operator. Thus, the FIGNN can capture more accurate correlations between conclusion nodes and ground nodes.

Next, we provide a more detailed and intuitive analysis of the inference process of FIGNN. We visualize the inference process in Fig. 1. In the zeroth iteration, the round ground

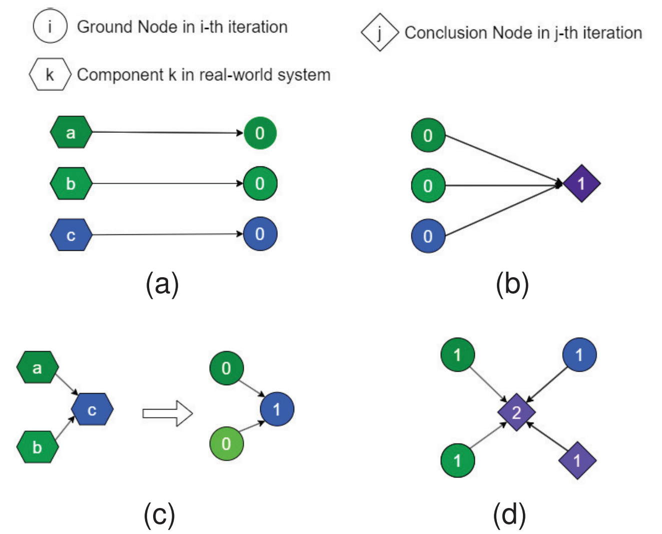

Fig. 1. Inference process in FIGNN. (a) Initialize of ground nodes. (b) Initialize of conclusion nodes. (c) Update of ground nodes. (d) Update of conclusion nodes.

nodes $V_g(0)$ are initialized by the fuzzify layer, as shown in Fig. 1(a), whereas rhombic conclusion nodes $V_c(0)$ remain $\mathbf{0}$. In the first iteration, $V_c(1)$ is initialized by $[V_g(0), \mathbf{0}]$, as shown in Fig. 1(b). Meanwhile, information aggregation among ground nodes updates $V_g(1)$ with attribute feature $V_g(0)$ in Fig. 1(c), which is the core advantage of graph-based methods. In the second iteration, $[V_g(1), V_c(1)]$ is used to update $V_c(2)$, as in Fig. 1(d), and $V_c(2)$ is the first effective conclusion nodes updated by both ground and conclusion nodes. Therefore, to achieve effective inference, FIGNN requires at least two iterations. In practice, the exact number of iterations needed for FIGNN depends on the characteristics and size of the problem. Obviously, FIGNN with too few iterations cannot handle complex problems, whereas FIGNN with too many iterations faces an over-smoothing problem. Unlike traditional GNNs, FIGNN learns the new node representation directly through inference, rather than simple averaging aggregation, which improves the robustness and reliability of fault feature extraction.

Stability is an important requirement for inference using an iterative approach, i.e., the output of graph inference should be limited to $[0, 1]$. Otherwise, this leads to the collapse of the inference process. The output of an inference network with T iterations is as follows:

$$V(T) = \widehat{A}^T V(0) \leq \widehat{A}^T \tag{13}$$

where $\widehat{A} = \mathbf{w} \odot A$, and the inequality holds due to the fact that fuzzified input $V(0) = [\mu(x), \mathbf{0}]$ is strictly limited to $[0, 1]$. Hence, the inference process on a graph is stable if only if $\widehat{A}^T$ is limited to $[0, 1]$, which induces the following condition for $\widehat{A}$:

$$\sum_{j=1}^{k} \widehat{A}_{ij} \leq 1, \ \forall\ i \in \{1, 2, \ldots, k\}. \tag{14}$$

Thus, normalizing the row vector $\mathbf{w}_i$ to ensure that $\sum_{j=1}^{k} \widehat{A}_{ij} = 1$ can ensure that the output of the network is delimited, and conclusion nodes made by V_c approximate 1 for fault samples. Considering $\mathbf{w}_{ij}$ should be positive, the normalization algorithm is defined as Algorithm 1.

Algorithm 1 Normalization Algorithm for $\widehat{A}$

Input: $\widehat{A}$, infinitesimal $\epsilon > 0$
Output: normalized $\widehat{A}$

for i in $[1, k]$ **do**
  $max = \max\left\{\widehat{A}_{ij}, j \in [1, k]\right\}$
  $min = \min\left\{\widehat{A}_{ij}, j \in [1, k]\right\}$
  for j in $[1, k]$ do
  $\widehat{A}_{ij} = \frac{\widehat{A}_{ij} - \min}{\max\text{-}\min + \epsilon}$
  end for
  end
  $\widehat{A_{i,:}} = \frac{\widehat{A_{i,:}}}{\sum_{j=1}^{k} \widehat{A_{ij}}}$
end for
return $\widehat{A}$

The next step for fuzzy inference is to compute the adaptation degree based on the rule premise and then normalize weights. The parameters in $\widehat{A}$ contain rules for all cases; therefore, different input features should activate different rules. In FIGNN, the adaptation degree mechanism changes the weights of information aggregation between nodes, which is similar to the attention mechanism. On the basis of this theory, we design Algorithm 2 to compute the adaptation degree.

Algorithm 2 Computation of Adaptation Degree

Input: normalized matrix,$\widehat{A}$ input vector x
Output: attention optimized$\widetilde{A}$

for i in $[1, k]$ **do**
  $\widehat{A_{i,:}} = \widehat{A_{i,:}} \odot x$
end for
end
$\widetilde{A}$ Algorithm 1 $(\widehat{A})$
return$\widehat{A}$

Because $\widehat{A_{i,:}} = \widehat{A_{i,:}} \odot x$ squeezes the weights, a re-normalized process is applied to the optimized attention weight matrix $\widehat{A}$ based on the above stability analysis, which truly realizes dynamic attention weights $\widetilde{A}$.

To improve the representation ability and scalability of FIGNN in complex practical problems with significant nonlinearity, an activation function should be applied to the output of each iteration. The sigmoid function is a widely used activation function in hidden layers of a neural network, which could ensure that the output is always between 0 and 1. However, the definition domain of the existing sigmoid function is in the real number field, which is inconsistent with the inference process. We therefore imitate the properties of the sigmoid function to design a suitable activation function for FIGNN

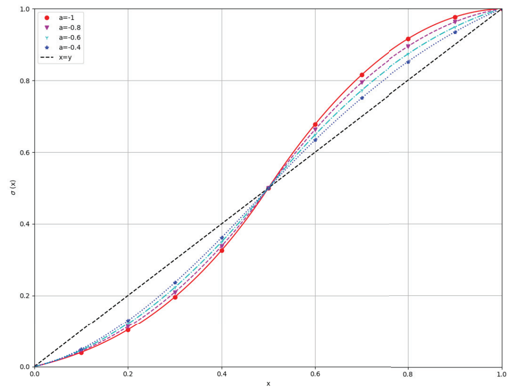

Fig. 2. Visualization of designed activation function $\sigma(x)$ with different a.

by imposing the following requirements:

$$\begin{cases} \sigma(x) = \begin{cases} f(x) & 0 \le x \le 0.5 \\ g(x) & 0.5 < x \le 1 \end{cases} \\ f(0) = 0, g(1) = 1 \\ f(0.5) = g(0.5) = 0.5 \\ \dot{f}(x) > 0, \dot{g}(x) > 0 \\ \ddot{f}(x) > 0, \ddot{g}(x) < 0 \\ \dot{f}(0) = \dot{g}(1) = \text{Const.} > 0 \end{cases} \tag{15}$$

where $\dot{f}(x)$ and $\dot{g}(x)$ are first-order derivatives of $f(x)$ and $g(x)$, respectively; $\ddot{f}(x)$ and $\ddot{g}(x)$ are second-order derivatives of $f(x)$ and $g(x)$, respectively; and Const. is a smaller constant.

To achieve these requirements, we can obtain

$$f(x) = \left(-\frac{26 * a}{7} - \frac{13}{5}\right)x^3 + \left(\frac{13 * a}{7} + \frac{27}{10}\right)x^2 + \frac{3}{10}x \tag{16}$$

$$g(x) = \left(-\frac{2 * a}{7} - \frac{1}{5}\right)x^3 + \left(\frac{17 * a}{7} + \frac{3}{10}\right)x^2 + \left(\frac{9}{10} - \frac{22 * a}{7}\right)x + a \tag{17}$$

where $a \in (-(21/20), -(21/65))$ is an adjustable parameter. Next, we visualize the designed activation function $\sigma(x)$ with different a values in Fig. 2. As shown in Fig. 2, the definition domain and output domain of $\sigma(x)$ are both $[0, 1]$, satisfying the requirements of inference process. An activation function $\sigma(x)$ with a larger a is close to a straight line, which can alleviate the problem of gradient vanishing but has weaker nonlinearity. Meanwhile, an activation function $\sigma(x)$ with a smaller a has strong nonlinear representation learning ability and can still alleviate gradient vanishing due to constraints $\dot{f}(0) = \dot{g}(1) = \text{Const.} > 0$.

C) The loss function of FIGNN and training strategy: The fault diagnosis problem is reduced to a semi-supervised classification task. In FIGNN, conclusion nodes are utilized to predict the probability of a particular fault occurring. Thus, the loss function of training is defined as follows:

$$\mathcal{L}_c = \frac{1}{n}\sum_{i=1}^{n} l\left(\left(V_c^i(T), Y^i\right); \mathbf{w}\right) \tag{18}$$

where $\mathbf{w}$ is a trainable weights matrix, $V_c(T)$ is the representation of conclusion nodes with T inference iterations, Y is the true label, and $l(\cdot)$ is the cross-entropy function.

The matrix multiplication and polynomial activation functions are easily differentiable. The derivative of the triangular membership function at the inflection point can be replaced by the average of the derivatives of the two sides. Thus, similar to other neural networks, the FIGNN can be effectively trained using a gradient descent approach.

B. Reliable Graph Structure Construction

According to the fuzzy inference theory, the edges of a graph in FIGNN constitute the rule set. Considering the connections between $[V_g, V_c]$, the adjacency matrix has the following block form:

$$A = \begin{bmatrix} \mathrm{IA} & \mathbf{0} \\ \mathrm{IF1} & \mathrm{IF2} \end{bmatrix} \tag{19}$$

where $\mathrm{IA} \in \mathbb{R}^{(f*d)*(f*d)}$ describes the rules of interactions between ground nodes corresponding to components and reflects the intrinsic topological relationships; $\mathrm{IF1} \in \mathbb{R}^{n*(f*d)}$ is an adjacency matrix between ground nodes and conclusion nodes, which describes the inference rules of components to the fault diagnosis; and $\mathrm{IF2} \in \mathbb{R}^{n\times n}$ describes the rules between conclusion nodes. A high-quality graph structure is crucial for accurate inference.

The graph construction problem can be formulated as an optimization problem, learning an optimal graph structure for the inference process. Assuming that the inference weight $\mathbf{w}$ can be well trained with a given A, then the graph construction problem can be described as an optimization problem

$$\min_{A\in\{0,1\}^{k\times k}} \frac{1}{n}\sum_{i=1}^{n} l\left(\left(V_c^i(T), Y^i\right); \mathbf{w}, A\right). \tag{20}$$

However, this optimization problem is difficult to solve using the gradient descent method due to the discrete $\{0, 1\}$ constraint in the graph structure. To address this issue, we revise it as a pruning problem with a sparsity threshold κ

$$\min_{A\in\{0,1\}^{k\times k}} \frac{1}{n}\sum_{i=1}^{n} l\left(V_c^i(T), Y^i; \mathbf{w}, A\right), \quad \text{s.t.}\|A\|_0 \le \kappa. \tag{21}$$

The graph structure describes local neighbor relationships; therefore, we introduce the sparsity constraint $\|A\|_0 \le \kappa$. The sparse graph structure can alleviate the over-smoothing problem of GNNs. In the pruning problem, the constraints $\|A\|_0 \le \kappa$ and $A \in \{0, 1\}^{k\times k}$ are easily satisfied, and pruning before training (PBT) methods are suitable and effective for the pruning problem (21).

PBT is a class of pruning methods that do not rely on pretrained weights. The core of PBT is to score the connections and decide whether to connect or not. PBT can be used to determine connection relationships without data by analyzing prior knowledge of the piping and instrument diagram (P&ID) in industrial systems. The P&ID representation focuses on the flow of the pipework and how the process is controlled, showing how the pipework system connects the industrial process equipment together, which is our desired interaction relationship. Inference relationships can also be determined using prior knowledge. However, knowledge-driven methods are time-consuming and not easily generalizable. Inspired by saliency-based PBT methods, we provide two available saliency scoring approaches for FIGNN, including similarity-driven saliency scoring and causal-analysis-driven saliency scoring.

Cosine similarity with normalization is a widely used similarity metric that incorporates vector normalization as a preprocessing step. The normalization step involves scaling the vectors to unit length, ensuring that the magnitude of each vector does not influence the similarity calculation. Given the nodes $V = [V_g; V_c]$, the cosine similarity between any pair of nodes is

$$\mathrm{score}_{\mathrm{IA}}(i, j) = \mathrm{sim}(i, j) = \frac{V^i}{\|V^i\|} \cdot \frac{V^j}{\|V^j\|}. \tag{22}$$

Similarity-driven saliency scoring can directly describe the interaction among ground nodes. However, the relationship between ground nodes and conclusion nodes cannot be scored directly by similarity. Consider that a component that performs significantly differently from normal can easily contribute to fault inference. The IF1 vector for fault i can be written as

$$\mathrm{score}_{\mathrm{IF1}}(:, i) = 1 - \mathrm{sim}\left(V_g^i, V_g^0\right). \tag{23}$$

Assigning these vectors to the corresponding rows of IF1 is to get the score of the IF1 block. The IF2 block in A can be constructed using knowledge because conclusion nodes are all human-assigned, and thus their relationships do not require quantitative analysis. Thus, the graph structure A can be constructed according to the score, i.e.,

$$A(i, j) = \begin{cases} 1, & |\mathrm{score}(i, j)| \ge \lambda \\ 0, & |\mathrm{score}(i, j)| < \lambda \end{cases} \tag{24}$$

where $\lambda > 0$ is a pre-given threshold.

Similarity is only a simple metric of relationships because the graph for inference should be a directed graph. Granger causality analysis is a statistical method used to determine the causal relationship between two time series variables. Assuming X and Y are two time series variables, X can be estimated by following two autoregressive (AR) models of order ord, i.e.,

$$X_t = c + \sum_{i=1}^{\mathrm{ord}} \phi_i X_{t-i} + \varepsilon_t \tag{25}$$

$$X_t' = c + \sum_{i=1}^{\mathrm{ord}} \phi_i X_{t-i} + \sum_{i=1}^{\mathrm{ord}} \theta_i Y_{t-i} + \varepsilon_t. \tag{26}$$

Then, we conduct hypothesis tests, such as the F-test, to determine whether the lagged values of variable Y significantly contribute to the prediction of the variable X. Two indicators in the F-test will be applied. The p-value describes the probability of the null hypothesis being accepted, which can be used to replace similarity score. The f-value describes the strength of causality, which can be used to initialize the weight.

C. Traceability of Diagnostic Result for FIGNN

Here, we design an interpreter based on reverse inquiry to show the main information spreading during the graph inference process. It is worth noticing that the interpreter only filters the FIGNN key inference process, and the produced interpretation is from the inference process. For ease of understanding, we provide an example to show graph inference. Assuming that we have a two-layer FIGNN with an initial value $[V_g(0), \mathbf{0}]$, the information aggregation process is as follows:

$$\begin{aligned} V(1) &= \left[V_g(1), V_c(1)\right] = \mathrm{FIGNN}\left(\left[V_g(0), \mathbf{0}\right]\right) \\ V(2) &= \left[V_g(2), V_c(2)\right] = \mathrm{FIGNN}\left(\left[V_g(1), V_c(1)\right]\right). \end{aligned} \tag{27}$$

As $V_c(2)$ can be treated as the final conclusion, the index of $V_c(2)[i] = \max(V_c(2))$ is selected. As $V_c(2)[i]$ is determined by $[V_g(1), V_c(1)]$, we can find top m nodes $\{V(1)[1], \ldots, V(1)[m]\}$ that contribute most to $V_c(2)[i]$. Here, m is a given positive parameter. Those nodes are connected with $V_c(2)[i]$. Then, we repeat this operation on the nodes $\{V(1)[1], \ldots, V(1)[m]\}$ to individually select the most relevant m nodes in $[V_g(0), \mathbf{0}]$. These selected nodes are activated, and their corresponding edges are characterized for inference in FIGNN.

Algorithm 3 Whole Process of FIGNN in Fault Diagnosis

Input: Normalized sample matrix $\mathbf{S}$, threshold λ.
Output: Diagnostic results.

Graph construction
Determine the number and properties of nodes.
Calculate saliency score among ground nodes.
if saliency score$(i, j) \geq \lambda$ **then**
  $IA(i, j) = 1$
end if
Calculate saliency score between ground nodes and conclusion nodes.
if saliency score$(i, j) \geq \lambda$ **then**
  $\mathrm{IF}_1(i, j) = 1$
end if
Manually construct IF_2.
Initialize parameter $\mathbf{w}$ by saliency score.
FIGNN forward propagation
$X = FuzzifyLayer(\mathbf{S})$
for t as iterations of FIGNN **do**
  Compute adaptation degree by Algorithm 2.
  Inference new X by (12).
  Activate X by (16) and (17).
end for
Train FIGNN using gradient descent approach.
Fault diagnosis with a trained FIGNN.

The overall fault diagnosis process of FIGNN is shown in Algorithm 3. To intuitively understand the FIGNN, its framework is shown in Fig. 3.

IV. Experiments

In this section, we evaluate the effectiveness of the proposed FIGNN in two cases: 1) the fault diagnosis of Tennessee Eastman (TE) process and 2) the fault diagnosis of three-phase

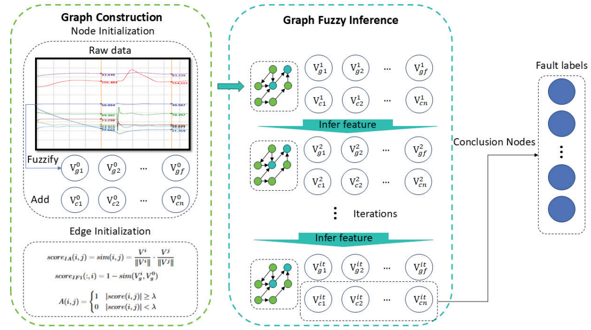

Fig. 3. Framework of the proposed FIGNN.

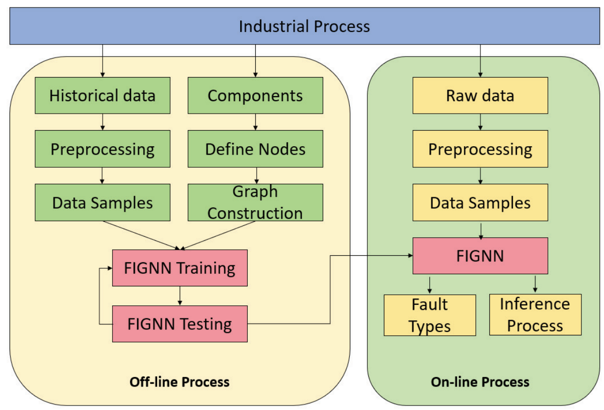

Fig. 4. Framework of fault diagnosis method based on FIGNN.

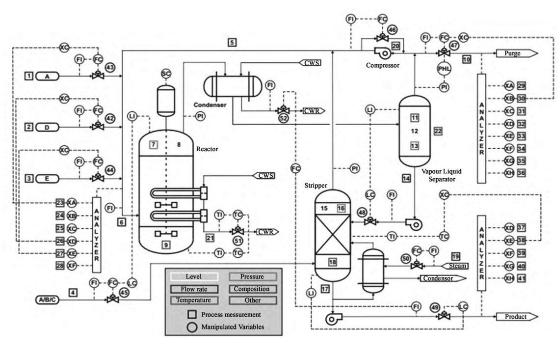

Fig. 5. Process and instrument diagram of process model for the TE process.

flow facility. In addition, the interpretability of diagnostic results for FIGNN is analyzed and discussed. The corresponding fault diagnosis method based on FIGNN is shown in Fig. 4. The data is segmented into samples after preprocessing, and the final diagnosis is derived from the FIGNN after fuzzification.

A. Experiments on the TE Process

1) Data Description and Preprocessing: The TE process is a complex chemical process with strong nonlinearity, and it is widely utilized as a benchmark dataset to validate the performance of fault detection and diagnostic approaches. The control strategy of the TE process is shown in Fig. 5. There are 41 measurements numbered from 1 to 41 and 11 manipulated

TABLE I
STATISTICS OF THE TE FAULTS

ID	Types	Disturbed Value
1	Step	A/C-ratio of stream 4, B composition constant
2	Step	B composition of stream 4, A/C-ratio constant
3	Step	D feed (stream 2) temperature
4	Step	Cooling water inlet temperature of reactor
5	Step	Cooling water inlet temperature of separator
6	Step	A feed loss (stream 1)
7	Step	C header pressure loss (stream 4)
8	Random	A/B/C composition of stream 4
9	Random	D feed (stream 2) temperature
10	Random	C feed (stream 4) temperature
11	Random	Cooling water inlet temperature of reactor
12	Random	Cooling water inlet temperature of separator
13	Drift	Reaction kinetics
14	Stiction	Cooling water outlet valve of reactor
15	Stiction	Cooling water outlet valve of separator
16	Random	(unknown); deviations of heat transfer within stripper
17	Random	(unknown); deviations of heat transfer within reactor
18	Random	(unknown); deviations of heat transfer within condenser
19	Stiction	(unknown); re-cycle valve of compressor, underflow separator (stream 10), underflow stripper (stream 11), and steam valve stripper
20	Random	(unknown)

variables numbered from 42 to 52, which are used to monitor and control the whole process. The original TE process includes 20 process disturbances shown in Table I, which could be utilized to comprehensively test the performance of the proposed method.

The TE process simulation is conducted, as suggested by a previous study [40], [41], where the process dataset can be downloaded from https://github.com/camaramm/tennessee-eastman-profBraatz. This dataset contains 21 training data files and 21 testing data files for each fault case and normal condition. Each training data file contains 24 960 numbers of data consisting of 24 h and each testing data file contains 49 920 numbers of data consisting of 48 h. The sample time is set to 3 min and 52 values are obtained for each sampling. An observation vector at a sample time is given as follows:

$$\mathbf{x} = [\mathrm{Xmeas}\,(1)\,, \mathrm{Xmeas}\,(2)\,, \ldots, \mathrm{Xmeas}\,(41)\,, \\ Xmv\,(1)\,, \ldots, Xmv\,(11)]^T \tag{28}$$

where Xmeas(i) is the ith measured variable and $Xmv(j)$is the jth manipulated variable. To satisfy the characteristics of the TE process, the following pre-processing methods are applied to the original data.

The sliding average filter is used to reduce the noise of the data matrix. It can effectively smooth the input signal by averaging its nearby data points, which helps to reduce high-frequency noises and variations. The sliding average filter is formulated as follows:

$$\mathbf{y}\,[t] = \frac{\mathbf{x}\,[t] + \mathbf{x}\,[t-1] + \cdots + \mathbf{x}\,[t-M+1]}{M} \tag{29}$$

where M is the sliding window size, which is set to 3 in our experiment. It acts as a low-pass filter, allowing slower changes in the input signal to pass through while attenuating rapid fluctuations.

Then, the filtered data are standardized by component as follows:

$$\bar{\mathbf{y}}\,[t] = \frac{\mathbf{y}\,[t] - \mathrm{mean}\,(\mathbf{y}_{\mathrm{normal}})}{\mathrm{std}\,(\mathbf{y}_{\mathrm{normal}})} \tag{30}$$

where std($\mathbf{y}_{\mathrm{normal}}$) denotes the standard deviation of $\mathbf{y}_{\mathrm{normal}}$, and $\mathbf{y}_{\mathrm{normal}}$ is the normal data of all components. After standardization, the scales of different components are unified. Inspired by fuzzy control, the rate of variable change is easily computed as follows:

$$\mathbf{d}\,[t] = \bar{\mathbf{y}}\,[t] - \bar{\mathbf{y}}\,[t-1]\,. \tag{31}$$

In a fuzzy control system, the error value of the system and the rate of change of the error are usually taken as the two inputs to the fuzzy controller. As the rate of change contains the necessary information for controlling, it is also useful for fault diagnosis. The ground nodes are generalized as $V_g = [\bar{\mathbf{y}}, \mathbf{d}]$. Then the preprocessed data are cut by a sliding window with a size of 50, achieving a certain degree of crossover between the samples.

2) Comparative Methods and Evaluation Metrics: To evaluate the superiority of our proposed model, we conduct a comparative experiment with DCNN [12], LSTM [17], BiGRU [42], PTCN [22], and TGGL [25]. For the proposed FIGNN, there are two versions, including FIGNN-S and FIGNN-C, where FIGNN-S constructs an adjacency matrix by similarity and FIGNN-C constructs an adjacency matrix by causality. The results of fault diagnosis are the following four scenarios, where positive and negative correspond to outputs, and true and false are the labels of samples.

Three metrics are chosen to comprehensively evaluate the performance of all fault diagnosis methods. ACR is the total classification accuracy of a fault diagnosis model considering all the testing samples. ACR is defined as follows:

$$\mathrm{ACR} = \frac{\sum N_{\mathrm{TP}}}{\sum N_{\mathrm{TP}} + \sum N_{\mathrm{FN}}} = \frac{\sum N_{\mathrm{TP}}}{N_{\mathrm{total}}}. \tag{32}$$

Another widely used criterion is the F1-score. The F1-score is the reconciling average of Precision and Recall metrics, which are more comprehensive and reliable. They are defined as follows:

$$\mathrm{Precision} = \frac{\sum N_{\mathrm{TP}}}{\sum N_{\mathrm{TP}} + \sum N_{\mathrm{FP}}} \tag{33}$$

$$\mathrm{Recall} = \frac{\sum N_{\mathrm{TP}}}{\sum N_{\mathrm{TP}} + \sum N_{\mathrm{FN}}} \tag{34}$$

$$F1 = \frac{2 * \mathrm{Precision} * \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{35}$$

To facilitate the analysis of interpretability, we define the density metric of the graph to measure its connectivity as follows:

$$\mathrm{Density} = \frac{N_{\mathrm{non-zero}}}{N} \tag{36}$$

where $N_{\mathrm{non-zero}}$ is the number of non-zero elements in the graph adjacency matrix, and N is the total number of elements in the graph adjacency matrix.

In our experiment, the ACR metric is utilized to intuitively evaluate the performance of all competitors. The Precision, F1-score, and Density are mainly used to comprehensively analyze the proposed FIGNN.

TABLE II
RESULTS OF FAULT DIAGNOSIS

Label	True	False
Positive	True Positive (TP)	False Positive (FP)
Negative	True Negative (TN)	False Negative (FN)

3) Experimental Setting: To ensure the consistency of training conditions, all versions of FIGNN are trained with 300 epochs. In every epoch, the dataset is divided into mini-batches with a size of 16 data samples. For the TE process, each component is described by four fuzzy sets: unusually high, unusually small, unusual rise, and unusual fall, resulting in 208 ground nodes corresponding to 52 components. The 21 conclusion nodes for classifying results and 19 fully connected intermediate conclusion nodes are applied for the task. A more curved activation function is required, so the hyperparameter a in the activation function $\sigma(x)$ is set to -1. The cross-entropy loss is chosen for the multiclassification problem with 21 conclusion nodes. An Adam optimizer with a learning rate of 0.05 is used to update the trainable parameters.

4) Diagnostic Results: The experimental results are shown in Table III, where original results of competitors are directly shown according to the primary literature. In Table III, we report the ACR results of each fault and their average value. It can be found from these results that our proposed FIGNN-C achieves the most superior average diagnostic accuracy with 98.10%. In addition, the improvement of FIGNN is considerably significant. For example, FIGNN-C improves around 4.20% in terms of the average ACR metric over the second-best PTCN method. For Fault 15, FIGNN-C outperforms the suboptimal BiGRU by 37.9%. These experimental observations consistently demonstrate the superiority of the proposed FIGNN method. And, under different faults, the proposed FIGNN-C method achieves the optimal performance over 15 faults, which further verifies its stability and robustness for the complex TE process.

Next, we also provide fine-grained analysis for FIGNN. Specifically, FIGNN-C consistently outperforms FIGNN-S except in a few cases (Faults 3, 8, 16, and 18). Moreover, FIGNN-C still outperforms the FIGNN-S with remarkable improvements on Faults 9 and 15 at 61.0% and 30.0%, respectively. Faults 9 and 15 are considered as the hardest fault group to detect because their abnormal information is weak. For existing GNN-based fault diagnosis methods, these tiny abnormalities are easily covered up by the surrounding normal data. For FIGNN, the inference relationship between nodes ensures that all anomalous information can be activated and extracted, thus preventing the disappearance of tiny abnormalities. Further, the inference function of FIGNN also yields potential anomalies that are canceled out by the controller, providing additional information for fault diagnosis.

To visually analyze the advantages of the proposed method, we visualize the confusion matrix of FIGNN on the test dataset in Fig. 6, which displays the distribution of predicted labels. Following the common labeling criteria of multiclassification tasks, the fuzzy result is defuzzified using the following rule:

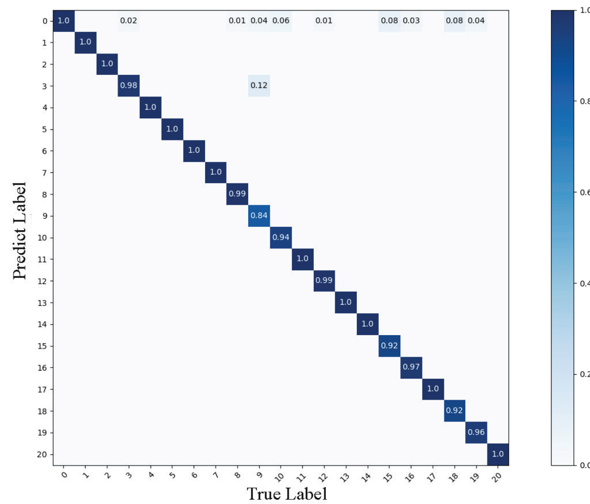

Fig. 6. Visualization of the confusion matrix for the proposed FIGNN on the TE process dataset.

if the value of a node is the largest among the 21 labeled nodes and exceeds 0.5, then the label of the node is defined as the final fault classification. As discussed in [12], Faults 3, 9, and 15 could hardly be distinguished from the normal state, especially for traditional fault detection methods. Based on the observations of the dataset, neither the fault location nor the reactor showed significant anomalies, with some unusual fluctuations only at the strip tower. Adjacent to the fault feature variables in a DCNN are usually irrelevant variables, and the convolution and pooling processes make the features disappear. RNN methods (LSTM and BiGRU) focus more on temporal transformations, and features are more easily preserved. For spatial graph convolution methods represented by undirected graphs (PTCN, TGGL), node features are averaged or weighted averages of themselves and the neighboring nodes, but not as bad as DCNN. Fortunately, our approach achieves an ideal performance on Fault 3. The lowest ACR of FIGNN occurs in Fault 9, which shares the same root cause in the D feed (Stream 2) temperature but a different fault type (random or step). Therefore, the majority of wrongly detected Fault 9 are recognized as Fault 3. Nevertheless, our method only misidentifies 4% of faulty samples as normal states for Fault 9, verifying the advantages of its fault identification. This is mainly attributed that the adaptation degree ensures that the message-passing is only activated by anomalous features. In addition, the directionality of the graph in FIGNN ensures that anomalous information can be efficiently propagated to the relevant nodes without being averaged out by normal signals.

5) Parameter Sensitivity Analysis: The graph construction involves crucial hyperparameters that need to be tuned. The similarity-based graph construction algorithm has a threshold parameter λ, whereas the causality-based algorithm involves two hyperparameters. Then, we show the changes of Density, Precision, ACR, and F1-score metrics with respect to λ, and the results of the knowledge-driven method are shown in Table IV. As shown in Table IV, the well-tuned similarity-based FIGNN is still slightly lower than the knowledge-driven

TABLE III
FAULT DIAGNOSIS RESULTS OF ALL COMPETITORS ON THE TE PROCESS DATASET. THE BEST RESULTS ARE BOLD AND UNDERLINED

Fault	DCNN	LSTM	BiGRU	PTCN	TGGL	FIGNN-S	FIGNN-C
Normal	0.9780	0.9400	0.9690	0.9924	**1.0000**	**1.0000**	**1.0000**
Fault 1	0.9860	**1.0000**	0.9860	0.9931	**1.0000**	**1.0000**	**1.0000**
Fault 2	0.9850	**1.0000**	0.9720	0.9819	**1.0000**	**1.0000**	**1.0000**
Fault 3	0.9170	0.7600	0.9350	0.8804	0.5200	**1.0000**	0.9800
Fault 4	0.9700	**1.0000**	0.9740	0.9956	**1.0000**	**1.0000**	**1.0000**
Fault 5	0.9150	**1.0000**	0.9980	0.9786	**1.0000**	**1.0000**	**1.0000**
Fault 6	0.9750	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
Fault 7	0.9990	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
Fault 8	0.9220	0.9900	0.7530	0.9160	0.9300	**1.0000**	0.9900
Fault 9	0.5840	0.1900	0.8070	0.6601	**0.8400**	0.2300	**0.8400**
Fault 10	0.9640	0.8900	**1.0000**	0.9276	0.8400	0.9400	0.9400
Fault 11	0.9840	0.9900	0.9650	0.9798	**1.0000**	0.9900	**1.0000**
Fault 12	0.9560	0.9900	0.9601	0.9704	**1.0000**	0.9700	0.9900
Fault 13	0.9570	0.9600	0.9530	0.8969	0.8400	**1.0000**	**1.0000**
Fault 14	0.9870	**1.0000**	0.9960	0.9964	**1.0000**	**1.0000**	**1.0000**
Fault 15	0.2800	0.0600	0.5410	0.0035	0.5200	0.6200	**0.9200**
Fault 16	0.4420	0.8600	0.7880	0.9685	0.9500	**0.9900**	0.9700
Fault 17	0.9450	0.9900	0.9700	0.9254	**1.0000**	**1.0000**	**1.0000**
Fault 18	0.9390	0.9700	0.9230	0.9049	**1.0000**	0.9900	0.9200
Fault 19	0.9860	0.7400	0.9260	0.9650	**1.0000**	0.9300	0.9600
Fault 20	0.9330	0.9900	0.9810	0.8825	**1.0000**	**1.0000**	**1.0000**
Average	0.8820	0.8730	0.9270	0.9392	0.9250	0.9250	**0.9810**

TABLE IV
PERFORMANCE RESULTS OF FIGNN-S WITH DIFFERENT THRESHOLDS

λ	0.01	0.1	0.2	0.5	0.7	0.9	knowledge
Density	0.4469	0.2638	0.2028	0.1537	0.1423	0.1115	0.1066
Precision	0.9978	0.9958	0.9826	0.9730	0.9664	0.8422	**0.9983**
ACR	0.7599	0.7675	0.7850	**0.9250**	0.9187	0.7552	0.9211
F1 score	0.8627	0.8668	0.8727	0.9483	0.9419	0.7963	**0.9581**

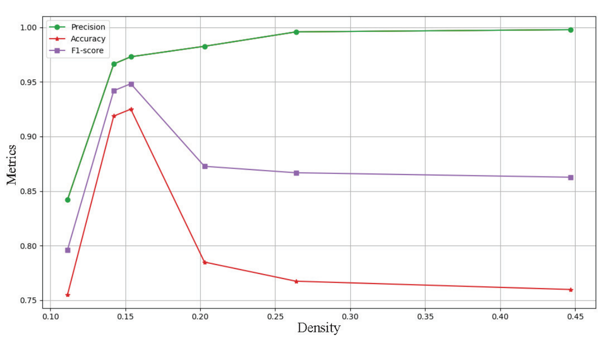

Fig. 7. Metrics change with different density settings.

TABLE V
PERFORMANCE RESULTS OF FIGNN-C WITH DIFFERENT P-VALUES IN THE INTERACTION AND INFERENCE BLOCKS

p in IA / p in IF	0.05 / 0.05	1e-7 / 0.05	1e-7 / 0.1	1e-7 / 0.5	similarity	knowledge
Density	0.2877	0.1908	0.2127	0.2281	0.1537	0.1066
Precision	0.9930	0.9444	0.9708	0.9756	0.9730	**0.9983**
ACR	0.6853	0.9232	**0.9584**	0.7683	0.9217	0.9211
F1 score	0.8109	0.9336	**0.9645**	0.8707	0.9466	0.9581

method, which demonstrates that the robustness and effectiveness of the similarity-based FIGNN method are weak. Thus, a causality-based graph construction strategy is necessary for FIGNN.

In the causality-based algorithm, the p-value is tuned separately in the interactive block IA, and the inference block IF = [IF1, IF2]. The edge distributions of the similarity-based experiment provide guidance on the experimental setting of the causality-based algorithm, so some extreme parameters like 1e^{-7} are also selected. The results of FIGNN-C with different interaction and inference thresholds are listed in Table V. And, for intuitive comparison, we also show the results of FIGNN with similarity and KGs. In causal analysis, $p \leq 0.05$ is the recommended threshold for judging whether causality exists. We notice that the performance of well-tuned causal-based FIGNN is better than that of the knowledge-driven FIGNN. In the case where knowledge is not complete, it is possible to use causal analysis to get higher-quality graph connections for FIGNN. However, the performance of FIGNN with both graph construction methods has the same interesting phenomenon. Taking the FIGNN-S as an example, we visualize the variation of Precision, ACR, and F1-score metrics with density. As shown in Fig. 7, as the density of the adjacency matrix decreases, the precision metric decreases, while the ACR and F1-score metrics first increase and then decrease.

To explain the above phenomenon, several concepts are introduced, including positive edges, neutral edges, and negative edges. Positive edges consist of neighborhood information, which is crucial for GNNs to perform comprehensive feature extraction and representation learning. As mentioned in solving the over-smoothing problem [43], a certain proportion of edges are randomly pruned during the training process, and these are defined as neutral edges. Neutral edges can bring irrelevant patterns and dilute the contribution of positive edges. Negative edges correspond to wrong connection relationships

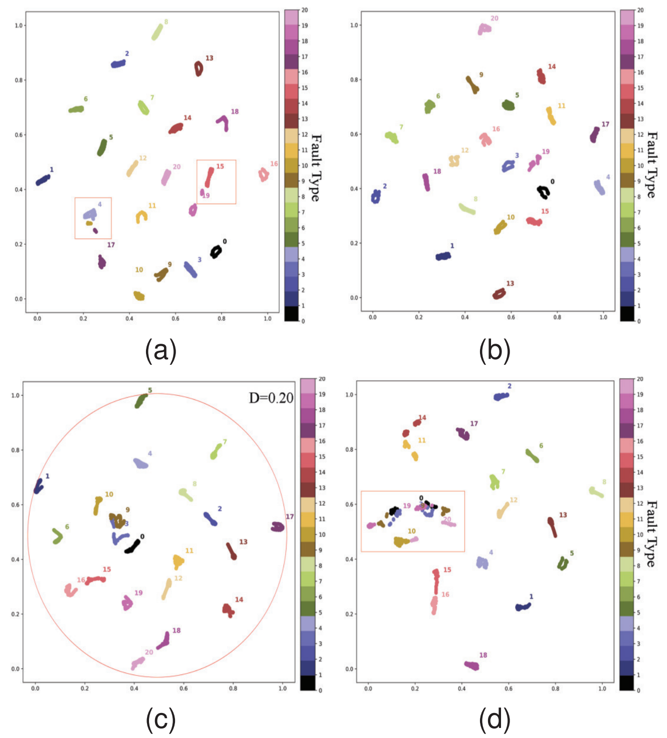

Fig. 8. t-SNE visualization of embedding representation of FIGNN at different densities of graph. (a) Density 0.1115. (b) Density 0.1537. (c) Density 0.2028. (d) Density 0.4469.

in a graph, resulting in a negative influence on information aggregation.

To show the impact of these edges more intuitively, we use t-SNE to visualize the embedding representation V_g of FIGNN at four different densities (0.1115, 0.1537, 0.2028, and 0.4469). First, positive edges are gradually introduced to provide a basis for differentiating individual faults. As shown in Fig. 8(a) with a density $D = 0.1115$, some faults are difficult to distinguish. For example, there is partial overlap in the samples of Faults 4 and 17, and the sample distributions of Faults 15 and 19 are close. This is mainly due to the fact that when the connection density is low, there are fewer positive edges in the graph, resulting in suboptimal information aggregation. In Fig. 8(b), when the density $D = 0.1537$, different faults have clear boundaries, and thus FIGNN achieves the best performance. However, due to all the saliency-based graph construction approaches cannot accurately distinguish between positive and neutral edges, some neutral edges are confused with positive edges when a decreasing saliency is gradually introduced. When the density comes to 0.2028 in Fig. 8(c), different from the cases with $D = 0.1115$ and $D = 0.1537$, the samples of normal state move from the margin to the center, which is mainly influenced by excessive neutral edges. More neutral edges will further dilute the proper fault characteristics. When the density is 0.4469, there are many neutral edges in the graph, diluting the contribution of the positive edges. Thus, as shown in Fig. 8(d), many faulty samples overlap with normal samples. Although FIGNN achieves close performance at densities of 0.2028 and 0.4469, there are significant differences in results distributions.

TABLE VI

PERFORMANCE WITH DIFFERENT ITERATIONS

Iterations	2	3	**4**	5
Precision	0.8971	0.9708	**0.9879**	0.9868
ACR	0.9472	0.9584	**0.9810**	0.9568
F1 score	0.9214	0.9645	**0.9844**	0.9715

The visualized experimental results are consistent with our theoretical interpretation of Fig. 7. This phenomenon occurs on all graphs constructed in this study, even though they are constructed in different ways. The difference between them is that the optimal points occur at different values of density. This indicates that a proper ratio of positive and neutral edges is the sign of an optimal case. Causality-based method FIGNN-C contains more interactions, which is the main reason for its better performance.

Then, given the optimal causal thresholds $[10^{-7}, 0.1]$, we show the impact of the number of iterations on the performance of FIGNN. The experimental results are shown in Table VI. In fact, the number of iterations in FIGNN is equivalent to the number of layers in GNNs. As shown in Table VI, FIGNN achieves the best performance when the number of iterations is 4. When there are relatively more iterations, FIGNN still maintains a satisfactory performance, indicating that FIGNN can alleviate the over-smoothing problem to a certain extent. This is mainly attributed to the fact that FIGNN could prune some useless edges according to causal relationships.

6) Pruning FIGNN by GLT: Here, we show the impact of pruning on the performance of FIGNN. An initial FIGNN model that has not been sufficiently trained is given, where there is a large number of redundant parameters. The lottery ticket hypothesis (LTH) [44] and its variant GLT [38] on the graph network are pruning after training (PAT) methods that can make the pruned network perform similar to or even better than the original network. The results of Precision, ACR, and F1-score metrics under different pruning conditions are shown in Table VII. Specifically, the results in the next column are obtained by performing the GLT on the network connections with the training results in the previous column.

In the initialization of A, connections are considered as knowledge-driven rules, and the training process corresponds to parameter pruning. The experimental results show that the data-driven PAT approach can maintain and even improve the performance of FIGNN by removing weak connections in A and retraining the network with the same initial value. The performance improvement suggests that connections in the adjacency matrix can be updated using a data-driven method. The case with a 0.11 density has a better ACR and a close F1-score compared with the knowledge-driven A. This suggests that after several iterations, the adjacency matrix constructed using data-driven pruning can be comparable to the adjacency matrix constructed by pure knowledge.

B. Experiments on the Three-Phase Flow Facility

Three flow facility (TFF) at Cranfield University provides a closed-loop system of pressurized pipelines with air, water,

TABLE VII

PERFORMANCE OF FIGNN WITH DIFFERENT PRUNING CONDITIONS. HERE, THE RATIO INDICATES THE NUMBER OF DENSITY AS A PERCENTAGE OF THE INITIAL NUMBER OF DENSITY

Density **Ratio**	0.175 100%	0.150 85.7%	0.140 80.0%	0.130 74.2%	0.120 68.5%	0.110 62.8%	0.100 57.1%	0.090 51.4%	0.080 45.7%	knowledge 60.9%
Precision	0.9423	0.9913	0.9896	0.9539	0.9788	0.9813	0.9657	0.9556	0.8865	**0.9983**
ACR	0.8753	0.9169	0.9137	0.8626	0.8881	**0.9265**	0.9009	0.8610	0.8738	0.9211
F1 score	0.9079	0.9526	0.9501	0.9059	0.9312	0.9531	0.9321	0.9058	0.8801	**0.9581**

TABLE VIII

FAULT DIAGNOSTIC RESULTS OF IAGNN, CTA-GCN, AND FIGNN ON THE TFF DATASET

Methods	**IAGNN**	**CTA-GCN**	**FIGNN-S**	**FIGNN-C**
ACR	0.8883	**0.9742**	0.9403	0.9659
F1 Score	0.9297	0.9735	0.9692	**0.9816**

and oil. TFF contains 24 measurements and 6 fault cases. The data used in the experiment could be downloaded from https://www.mathworks.com/matlabcentral/fileexchange/50938-a-benchmark-case-for-statistic-process-monitoring-cranfield-multiphase-flow-facility. Similar to the TE process, we use the same preprocessing method. We conduct a comparative study with two GNN approaches: IAGNN [24] and CTA-GCN [31]. The experimental results are shown in Table VIII, where some results directly come from the primary literature.

As shown in Table VIII, multihead attention-based IAGNN does not consider inner physical mechanisms during the graph construction process, which results in poor performance. However, FIGNN and CTA-GCN focus more on the quality of spatial relationship modeling, which is significantly crucial for the non-Euclidean characteristic of GNNs. Therefore, these two causality-guided approaches achieve the top two performances in F1-score metric. The experimental results on the two datasets comprehensively demonstrate the feasibility and effectiveness of the proposed FIGNN.

C. Visualization of the Fault Diagnosis Process for FIGNN

To show the traceability and interpretability of diagnostic results for FIGNN, a program that can automatically translate important inference processes of FIGNN into a flowchart formed interpretation is applied. Taking Fault 4 in the TE process as an example, we visualize it in Fig. 9. The root cause of Fault 4 is cooling water inlet temperature of reactor. Thus, the main fault behavior occurs in the valve position of cooling water outlet 51 and the temperature of cooling water outlet 21. The derivative of temperature 61 is inferred from the valve movement, which is a potentially anomalous behavior that is eliminated by the corresponding controller.

To demonstrate the advantage in traceability and interpretability of fault diagnostic result, the FIGNN is compared with a representative explainable AI (XAI) method, i.e., Local Interpretable Model-agnostic Explanation (LIME), where the LIME explainer is applied on FIGNN. The LIME model is downloaded from https://pypi.org/project/lime/ and training setting is consistent with FIGNN. Fault 4 is also selected

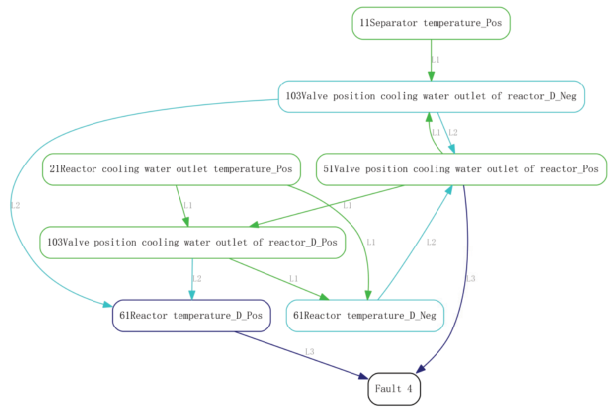

Fig. 9. Traceability of diagnostic result of FIGNN for Fault 4 on the TE process.

as the instance of LIME explainer and the result is shown in Fig. 10.

Observing the experimental results from Figs. 9 and 10, some interesting phenomena can be observed.

1) LIME produces lower quality local interpretations in Fault 4 than FIGNN. Specifically, in top 10 features, the first variable "Reactor cooling water outlet temperature" is considered to be the irrelevant variable for Fault 4, which is contrary to the cause of Fault 4. The second variable "Concentration of A in Reactor feed" is considered to be the most important variable to identify Fault 4, but it is actually unrelated to Fault 4. The third variable "Valve position cooling water outlet of reactor" is one of the causes of the Fault 4. Although the LIME achieves the appropriate judgment, it does not identify this variable as a priority influencing factor.
2) Interestingly, the proposed FIGNN establishes direct correlations between the conclusion node of Fault 4 and the ground nodes closely related to temperature variables, which is consistent with prior domain knowledge and causes of Fault 4.

The difference in interpretability between the FIGNN and LIME methods is mainly attributed to the following aspects.

1) The XAI methods represented by LIME mainly seek features that contribute significantly to the label, rather than analyzing the correlations between feature variables and fault conclusions. However, the FIGNN innovatively introduces conclusion nodes to represent fault types and establishes and explores associations between conclusion nodes and ground nodes corresponding to

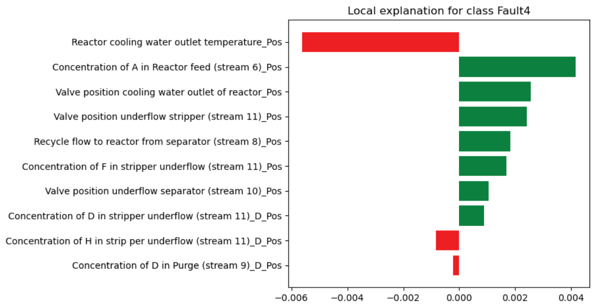

Fig. 10. Explanation of diagnostic result of LIME for Fault 4 on the TE process.

TABLE IX
FAULT DIAGNOSTIC RESULTS AND FIDELITY METRIC OF FIGNN AND GNNS WITH MLP CLASSIFIER UNDER NOISE

Methods	SNR	ACR	F1 Score	Fidelity
GNNs	5	0.7944	0.8104	6.7069
FIGNN	5	0.9430	0.9635	2.0467
GNNs	10	0.9083	0.9270	2.1722
FIGNN	10	0.9680	0.9760	0.7333
GNNs	20	0.9356	0.9544	0.6367
FIGNN	20	0.9705	0.9786	0.2143
GNNs	50	0.9365	0.9549	0.2447
FIGNN	50	0.9750	0.9814	0.1043

feature variables, which can help capture more relevant information about faults diagnosis process.

2) The training process of FIGNN is constrained by graph, which allows FIGNN to extract high-quality and causal features than explanatory machine learning models trained by XAI methods.
3) The parameters sharing in the IA block $V_g(t) = (\mathbf{w} \odot A)V_g(t-1)$, which is trained jointly with different related faults, provide global knowledge and explanations of the system while XAI methods are still limited to a local or small area of the sample.

To quantify the advantages of FIGNN in terms of reliability and interpretability, infidelity [45] is introduced. If a network $f(x)$ has good reliability and interpretability, perturbations on the samples $(x - I)$ should have a low impact on the network performance. Based on this theory, infidelity metric is defined as

$$\begin{aligned}&\mathrm{INFD}\,(\Phi, f, x)\\&= E_{I\in\mu_I}\left[\left(I^T\Phi\,(f, x) - (f\,(x) - f\,(x - I))\right)^2\right]\end{aligned} \tag{37}$$

where $I^T\Phi(f, x) = f(x) - f(x-I)$, the perturbation I is a random variable, and μ is the distribution of I. They also point out that applying random noise to the sample and computing fidelity is a robust method. We therefore add random noise with different signal-to-noise ratios (SNRs) and test the performance of FIGNN and classic GNN methods with an MLP classifier, respectively. The diagnostic performance and fidelity metric are reported in Table IX. The experimental results show that the performance and the fidelity metric of the FIGNN method are better than those of the GNNs with an MLP classifier under the influence of noise, indicating that the robustness and interpretability of FIGNN are indeed superior.

TABLE X
ABLATION STUDY OF FIGNN ON THE TE DATASET

Methods	Relu	w/o Adaptation degree	w/o ANFIS	FIGNN
ACR	0.7492	0.6134	0.8740	0.9810
F1 Score	0.7858	0.5626	0.9049	0.9844

D. Ablation Study

To verify the necessity and importance of key modules in FIGNN, this section conducts ablation experiments. In this ablation study, the crucial components (including new activation function, the updating of adaptation degree, and ANFIS) are removed from the complete FIGNN. The experimental results of FIGNN and corresponding degraded models are shown in Table X. From the results shown in Table X, the performance of FIGNN drops when the above components are removed, demonstrating their effectiveness and necessity. The reason is that the new activation function and adaptation degree could ensure the quality of fuzzy inference, where the adaptation degree in FIGNN is capable of activating accurate fuzzy rules.

V. CONCLUSION

To improve the representation learning, diagnostic analysis, and physical interpretability capabilities of the existing GNN-based fault diagnosis approaches, this paper proposed a novel fuzzy inference-guided GNN (FIGNN). The FIGNN redefined the information aggregation of GNN according to the Takagi–Sugeno fuzzy inference process, which promotes the interaction and transmission of logical knowledge embedded in system components. Moreover, to learn more reliable and interpretable connection relationships, a causality-analysis-based graph construction method was provided. Experiments on the TE and TFF datasets showed that the FIGNN model achieves satisfactory results in fault diagnosis problems, highlighting its strong traceability and interpretability of diagnostic results through an inference process. In the future, future research would parameterize signal temporal logic to achieve better extraction of temporal information.

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Youqing Wang (Senior Member, IEEE) received the B.S. degree in mathematics from Shandong University, Jinan, Shandong, China, in 2003, and the Ph.D. degree in control science and engineering from Tsinghua University, Beijing, China, in 2008.

He worked chronologically at The Hong Kong University of Science and Technology, Hong Kong, China; the University of California at Santa Barbara, Santa Barbara, CA, USA; the University of Alberta, Edmonton, AB, Canada; Shandong University of Science and Technology, Qingdao, Shandong; and the City University of Hong Kong, Hong Kong. He is currently a Professor with Beijing University of Chemical Technology, Beijing. His research interests include industrial data analysis, machine learning, fault diagnosis, fault-tolerant control, and iterative learning control for chemical and biomedical processes.

Dr. Wang was a recipient of several honors and awards, including the IET Fellow, the NSFC Distinguished Young Scientists Fund, the Journal of Process Control Survey Paper Prize, and the ADCHEM2015 Young Author Prize.

Tengxiao Yin is currently pursuing the master's degree in control science and engineering with Beijing University of Chemical Technology, Beijing, China.

His research interests include deep learning, graph neural networks, expert systems, and fault diagnosis of chemical systems.

Shenggang Zhu received M.S. degree in economics from Capital University of Economics and Business, Beijing, China.

He is a Senior Economist and the Vice President of China Environmental Association. His research interests include new energy, power systems, technology, and environmental protection. He participated in the national research and development project of high-power lithium-ion battery energy storage technology and the construction of the National Energy Administration Industrial Control System and Information Security Research and Development Center.

Xin Ma received the B.S. degree in biomedical engineering from Hebei University of Technology, Tianjin, China, in 2017, and the Ph.D. degree in control science and engineering from Beijing University of Chemical Technology, Beijing, China, in 2022.

From August 2022 to February 2023, she was an Exchange Student at the University of Surrey, Guildford, U.K. She is currently a Post-Doctoral Fellow with the College of Information Science and Technology, Beijing University of Chemical Technology. Her research interests include industrial data analysis, statistical learning, and intelligent fault diagnosis.

Li Liang received the Ph.D. degree in control science and engineering from Beijing Institute of Technology, Beijing, China, in 2020.

She is currently an Associate Professor with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing. Her research interests include differential games, multiagent systems, multiobjective optimization, and decision-making.

Jipeng Guo (Member, IEEE) received the Ph.D. degree in control science and engineering from Beijing University of Technology, Beijing, China, in 2023.

He is currently a Lecturer with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing. He has published several papers in top journals and conferences, such as IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Transactions on Image Processing (TIP), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), *ACM Transactions on Knowledge Discovery from Data* (TKDD), *Neural Networks*, AAAI, and ICASSP. His current research interests include computer vision, pattern recognition, machine learning, and clustering methods.